

RoPE attention is an exact forward-pass gradient step with softmax intact.

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Contributions: exact inference-time gradient identity and tokenwise decomposition

We express RoPE-softmax attention as an exact gradient step for arbitrary projection weights and affine biases. The construction places the head’s forward pass in effective-weight coordinates and resolves its query-dependent write into cached-token contributions. Full softmax and rotary position structure remain intact throughout.

An explicit forward-pass gradient step. Use the augmented input u_i and projections of section 3; write $L_i = \widehat{W}_q R(p_i)^\top / \tau$, $a_{ij} = L_i \widetilde{k}_j^\top$, and $c_{ij} = \rho(s_{ij}) / Z_i > 0$. For $B \in \mathbb{R}^{(m+1) \times r}$ define

$$\begin{aligned} \mathcal{E}_i(B) &= \frac{1}{2} \sum_{j \in \mathcal{I}_i} c_{ij} \|a_{ij}^\top B - (v_j - \mu_i)\|_2^2, \\ B_i^{(1)} &= 0 - \nabla_B \mathcal{E}_i(0) = \sum_{j \in \mathcal{I}_i} c_{ij} a_{ij} (v_j - \mu_i) = \Delta M_i, \quad y_i = \mu_i + u_i^\top B_i^{(1)}. \end{aligned} \tag{1}$$

Differentiation acts on B , with features, targets, and coefficients held fixed. Each query specifies the positive regression weights and reads out the resulting unit step from zero.

Exact tokenwise residual. At a fixed cache and query position, write $a_j = a_{ij}$ and choose a reference query 0. The adaptive write equals the reference write plus

$$e_i = u_i^\top (\Delta M_i - \Delta M_0) = \sum_{j \in \mathcal{I}} (c_{ij} - c_{0j}) (u_i^\top a_j) (v_j - \mu). \tag{2}$$

Each summand identifies a cached token’s contribution to the correction. Thus $y_i = \mu + u_i^\top \Delta M_0 + e_i$, and $e_i W_o$ is the exact projected correction.

RoPE as a bank of positional derivatives. For fixed content z_0 , $z(p) = R(p)z_0 = e^{pA}z_0$ satisfies $\partial_p^n z(p) = A^n z(p)$ for $n \geq 1$, and $z(p + \delta) - z(p) = (e^{\delta A} - I)z(p)$. These derivatives and finite differences are directly evaluable on frozen features. Our framework connects this derivative-operator representation to the exact full-softmax effective matrix. RollPE relates a special RoPE spectrum to Fourier differentiation [6].

Shared cache, query-conditioned weights. At a fixed cache and position, $\Delta M_i = L T_i$: $T_i \in \mathbb{R}^{d \times r}$ combines cached outer products with positive coefficients c_{ij} , and $\text{rank}(\Delta M_i W_o) \leq d$ in the original gauge. The mean is shared. The matrix gives a secant representation; differentiating the input and the coefficients gives the query Jacobian.

Equivalent matrices, one output map. For any fixed $g \in \mathbb{R}^{m+1}$, set $\sigma_i = u_i^\top g$. Then

$$\begin{aligned} a_{ij}^g &= a_{ij} - g, \quad s_{ij}^g = s_{ij} - \sigma_i, \quad c_{ij}^g = \frac{e^{\sigma_i} \rho(s_{ij} - \sigma_i)}{Z_i}, \\ \Delta M_i^g &= \sum_j c_{ij}^g a_{ij}^g (v_j - \mu_i), \quad y_i = \mu_i + u_i^\top \Delta M_i^g. \end{aligned} \tag{3}$$

These transformations preserve the probabilities and output. Key shifts give the subfamily $g = L_i b$, of dimension at most d ; general g can add one direction. We report matrices in the original-score gauge and evaluate transfer in output coordinates.

Connection to linear-attention gradient constructions. Von Oswald’s Proposition 1 [1] realizes a least-squares gradient step with linear attention; related fast-weight [2] and implicit-gradient [3] constructions use linear attention or linearized softmax. Setting $R = I$ and taking $c_{ij} \rightarrow 1/N_i$, with centered regression tokens, his projection/readout choices, and matched scale, recovers his step. Equation (1) supplies the exact full-softmax coefficients.

Pretrained realization. On Qwen2.5-0.5B, layer 23, all 240 context/head reconstruction rows pass. The combined prefix and full-attention writes reconstruct at float64 MSE 9.78×10^{-30} and 9.62×10^{-30} , respectively.

Mathematical construction

1. RoPE and the score in the original weights. Let $x_i \in \mathbb{R}^m$ be a column input to one head, with $W_q, W_k \in \mathbb{R}^{m \times d}$ and $W_v \in \mathbb{R}^{m \times r}$. Projected vectors are rows: $q_i = x_i^\top W_q$, $k_j = x_j^\top W_k$, and $v_j = x_j^\top W_v$. Set $\tau = \sqrt{d}$. For standard RoPE with fixed frequencies [4],

$$R(p) = e^{pA}, \quad A = \text{diag}_{\ell=1}^{d/2} \begin{pmatrix} 0 & -\omega_\ell \\ \omega_\ell & 0 \end{pmatrix}, \quad R(p_i)^\top R(p_j) = R(p_j - p_i). \quad (4)$$

With $\tilde{q}_i = q_i R(p_i)^\top$ and $\tilde{k}_j = k_j R(p_j)^\top$, define

$$a_{ij} = \frac{W_q R(p_j - p_i) W_k^\top x_j}{\tau} \in \mathbb{R}^m, \quad s_{ij} = \frac{\tilde{q}_i \tilde{k}_j^\top}{\tau} = x_i^\top a_{ij}. \quad (5)$$

2. Exact factorization through softmax. For deterministic attention, let $\mathcal{I}_i$ be a nonempty set of permitted keys; causal masking selects this set. With the scores in (5), write $N_i = |\mathcal{I}_i|$ and

$$Z_i = \sum_{j \in \mathcal{I}_i} e^{s_{ij}}, \quad \mu_i = \frac{1}{N_i} \sum_{j \in \mathcal{I}_i} v_j, \quad y_i = \frac{1}{Z_i} \sum_{j \in \mathcal{I}_i} e^{s_{ij}} v_j. \quad (6)$$

Here μ_i is the *uniform arithmetic mean* of permitted values. The divided difference

$$\rho(s) = \int_0^1 e^{ts} dt, \quad \rho(s) = \frac{e^s - 1}{s} \quad (s \neq 0), \quad \rho(0) = 1, \quad e^s - 1 = s\rho(s). \quad (7)$$

gives, for every query,

$$\boxed{y_i = \mu_i + x_i^\top \Delta M_i, \quad \Delta M_i = \frac{1}{\tau Z_i} \sum_{j \in \mathcal{I}_i} \rho(s_{ij}) [W_q R(p_j - p_i) W_k^\top x_j] [x_j^\top W_v - \mu_i]}. \quad (8)$$

Proof. The centered values sum to zero. With all sums over $\mathcal{I}_i$,

$$y_i - \mu_i = \frac{\sum_j (e^{s_{ij}} - 1)(v_j - \mu_i)}{Z_i} = \frac{\sum_j s_{ij} \rho(s_{ij})(v_j - \mu_i)}{Z_i} = x_i^\top \Delta M_i. \quad \square$$

Setting $R = I$ gives the position-free construction; RoPE supplies relative-position structure.

Convention. Reported matrices use the *original-score gauge* of (8); gauge changes transform both features and coefficients, as in (3).

Transformed scores, additive biases, and QK normalization. For pointwise ϕ with $\phi(0) = 0$, use $Z = \sum_j e^{\phi(s_{ij})}$ and $\rho_\phi(s) = (e^{\phi(s)} - 1)/s$ for $s \neq 0$; at zero, any finite value works (continuously, $\phi'(0)$ when

it exists). Sign-preserving soft-caps preserve nonnegative regression weights. Additive input-independent biases, including ALiBi, are absorbed by $(x_i^\top, 1)(a_{ij}^\top, b_{ij}^\top)^\top$. For nonlinear QK normalization (Qwen3 [5]), use the normalized, rotated query:

$$y_i = \mu_i + \tilde{q}_i S_i, \quad S_i = \frac{1}{\tau} \sum_j \frac{\rho(s_{ij})}{Z_i} \tilde{k}_j^\top (v_j - \mu_i). \quad (9)$$

Finite-precision evaluation in the declared gauge. With $M_i = \max_j s_{ij}$, $\bar{Z}_i = \sum_j e^{s_{ij} - M_i}$ and $\ell_i = M_i + \log \bar{Z}_i$,

$$c_{ij} = \frac{P_{ij} - e^{-M_i}/\bar{Z}_i}{s_{ij}} = \frac{P_{ij} - e^{-\ell_i}}{s_{ij}} \quad (s_{ij} \neq 0), \quad c_{ij} = e^{-\ell_i} \quad (s_{ij} = 0). \quad (10)$$

Evaluate near-zero scores with $e^{-\ell_i} \expm1(s_{ij})/s_{ij}$ and use log-domain arithmetic elsewhere. Numerical scaling preserves the original gauge.

3. Effective residual weights and affine projections. With $W_o \in \mathbb{R}^{r \times m}$ and residual input x_i , equation (8) gives

$$x_i^\top + y_i W_o = x_i^\top (I_m + \Delta M_i W_o) + \mu_i W_o. \quad (11)$$

For projection biases, use $u_j = (x_j^\top, 1)^\top$ and $\widehat{W} = [W^\top \ b]^\top$; the proof is unchanged. For pre-normalization, $x_i = \text{Norm}(h_i)$ and the residual is $h_i^\top + (\mu_i + u_i^\top \Delta M_i^{\text{aug}}) W_o$. Add output bias separately and sum projected heads, each with its assigned K/V bank.

4. Exact reference decomposition and matrix transfer. Fix a context bank and query position. For anchor 0, set $y_i^{\text{fr}} = \mu + x_i^\top \Delta M_0$. Then

$$\boxed{\begin{aligned} y_i - y_i^{\text{fr}} &= x_i^\top (\Delta M_i - \Delta M_0), \\ \|y_i - y_i^{\text{fr}}\|_2 &\leq \|x_i\|_2 \|\Delta M_i - \Delta M_0\|_{\text{op}}. \end{aligned}} \quad (12)$$

The contraction in (12) is the exact transfer residual. With $L = W_q R(p)^\top / \tau$ and $c_{ij} = \rho(s_{ij})/Z_i$,

$$\Delta M_i - \Delta M_0 = L \sum_{j \in \mathcal{I}} (c_{ij} - c_{0j}) \tilde{k}_j^\top (v_j - \mu). \quad (13)$$

These equations hold in augmented coordinates. Reference write plus coefficient refresh gives the exact readout. Freezing omits $x_i^\top (\Delta M_i - \Delta M_0)$; equation (2) resolves that residual by cached token.

5. Fixed-prefix and full-attention readouts. A fixed-prefix probe excludes the query’s own key/value to keep one common bank. If full attention adds a self value v_i^{self} with softmax mass π_i , then

$$y_i^{\text{full}} = (1 - \pi_i) y_i^{\text{prefix}} + \pi_i v_i^{\text{self}}. \quad (14)$$

The full-bank matrix uses its own scores and mean $(N\mu + v_i^{\text{self}})/(N+1)$. A changing bank adds $\mu_i - \mu_0$ to the error when the offset is also frozen.

6. Pretrained validation and frozen reuse. Qwen2.5-0.5B, layer 23 (zero-indexed): $C = 16$ prefixes of 511 tokens, all 14 heads, $Q = 32$ continuations at one position. Anchor $a = 0$ is each article’s observed next token; the others are the model’s 31 highest-ranked distinct nonspecial continuations. Each reads the same prefix-only bank.

Exact reconstruction. All 240 context/head reconstruction rows pass. Comparison with the native FP32 write gives MSE 7.01×10^{-14} and maximum absolute error 1.4×10^{-5} .

Frozen reuse. For each prefix, reuse the anchor’s matrix across its other queries. Let $o_{ci} = \sum_h y_{cih} W_{oh} \in \mathbb{R}^D$ denote the projected attention write.

$$E(f) = \frac{1}{C} \sum_{c=1}^C \frac{1}{(Q-1)D} \sum_{i \neq a} \|f_{ci} - o_{ci}\|_2^2, \quad \text{rRMSE}(f) = \sqrt{\frac{E(f)}{E(o_{ca})}}. \quad (15)$$

Pool squared errors with equal context weights and take ratios after pooling. All-anchor evaluation also averages over $a = 0, \dots, Q-1$ before taking ratios. Sum projected heads before squaring. Head-level baselines are

$$y_i^\mu = \mu, \quad y_i^{\text{lin}} = \mu + \frac{1}{N} \sum_j s_{ij}(v_j - \mu). \quad (16)$$

The latter is the first-order softmax expansion at zero logits with unit scale. Both undergo the same output projection and aggregation as the exact head.

Pooled MSE of the combined projected write:

Predictor	Anchor 0	All anchors
Constant anchor output	0.047787	0.045672
Cache mean ($\Delta M = 0$)	0.124573	0.124513
Frozen anchor matrix	0.324970	0.356763
Linearized softmax	0.161842	0.162068

Target RMS is 0.670545 for anchor 0 and 0.670824 across all anchors. Frozen rRMSE is 2.608 against anchor 0 and 1.615 against the cache mean; the all-anchor rRMSE is 2.795. In the all-anchor evaluation, thirteen individual heads have rRMSE 0.728–0.933 against their constant-anchor baselines.

References

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